What is X?

X represents the input features. In your project, it contains the email text.

Example:

- Congratulations! You won a prize.
- Meeting scheduled for tomorrow.

What is y?

y represents the target variable or output.

What is TF-IDF?

TF-IDF stands for:

Term Frequency – Inverse Document Frequency

It converts text into numerical values so machine learning algorithms can understand it.

t divides the dataset into two parts.

Dataset	**Percentage**	**Purpose**
Training data	80%	Train the model
Testing data	20%	Evaluate the model

Your dataset has 5,172 emails.

- Training: approximately 4,137 emails.
- Testing: 1,035 emails.

- NN is a supervised machine learning algorithm.
- It classifies a new data point based on the classes of its nearest neighbours.
-
- SVM is a supervised machine learning algorithm used for classification.
- Its main objective is to find the best decision boundary (hyperplane) that separates different classes.
- For your project, it separates spam emails from genuine emails based on their TF-IDF features.

Confusion matrix

Your KNN confusion matrix:

[[715 20]

[29 271]]

Your SVM confusion matrix:

[[725 10]

[2 298]]

Let's understand the four values.

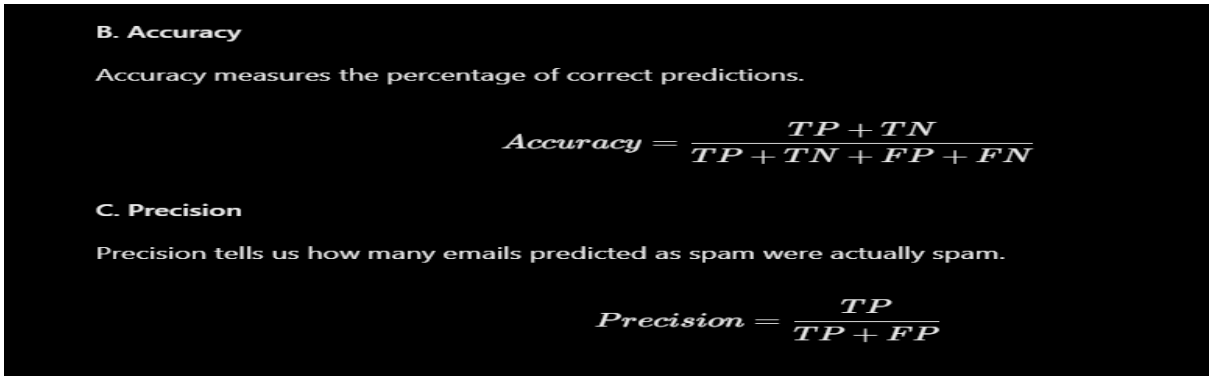
Term	Meaning
True Positive (TP)	Spam correctly predicted as spam
True Negative (TN)	Ham correctly predicted as ham
False Positive (FP)	Ham incorrectly predicted as spam
False Negative (FN)	Spam incorrectly predicted as ham

For your SVM model:

Actual / Predicted	Not Spam	Spam
Not Spam	725	10
Spam	2	298

This means:

- 725 genuine emails were correctly identified.
- 10 genuine emails were incorrectly identified as spam.
- 2 spam emails were incorrectly identified as genuine.
- 298 spam emails were correctly identified.



My project title is Email Spam Detection using Machine Learning.

The main objective of this project is to identify whether an email is spam or genuine.

First, I collected an email dataset in CSV format. Then I used Pandas to load and process the dataset.

I separated the email text and labels. I converted the labels into numerical values, where 0 represents Ham and 1 represents Spam.

After that, I used TF-IDF vectorization to convert the email text into numerical features.

I divided the dataset into 80% training data and 20% testing data.

Then I trained two machine learning algorithms: KNN and SVM.

Finally, I evaluated both models using accuracy, precision, recall, F1-score and confusion matrix.

My SVM model achieved 98.84% accuracy, while KNN achieved 95.27%.

Therefore, SVM performed better in my experiment